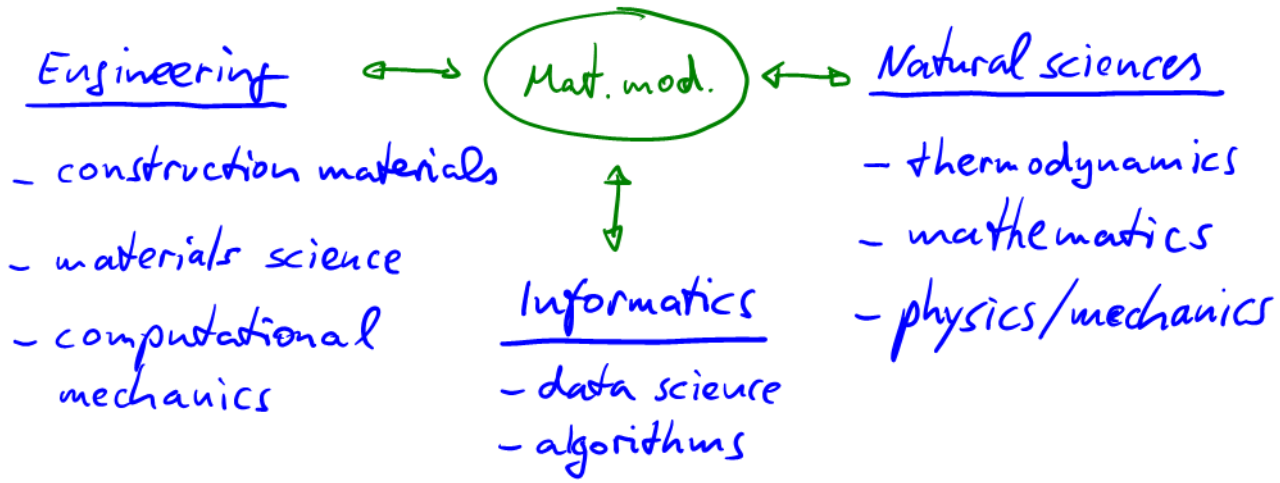
Mechanical Modeling of Materials

Why modeling of materials ?

- Improved understanding of material respone
  - ⇝ analysis of situations, which are difficult in the lab
- Analysis of stresses when knowing strains
  - ⇝ material law $\sigma := \sigma(\varepsilon)$
- Statically undetermined systems
  - ⇝ material laws provide additional equations
- Design of engineering structures
  - ⇝ strongly simplified (thin, homogeneous, linear elastic) structures → "engineering mechanics"
  - ⇝ arbitrary (e.g. inelastic, thick, complex-shaped) structures → computational simulation (e.g. FEM)

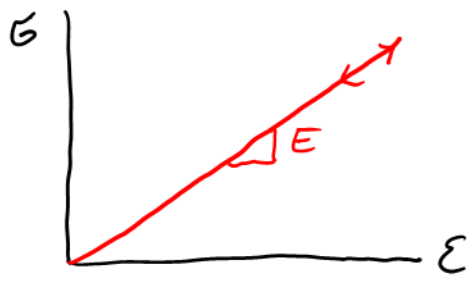
  ⇒ requires predictive material models in 3D

Material modeling is interdisciplinary



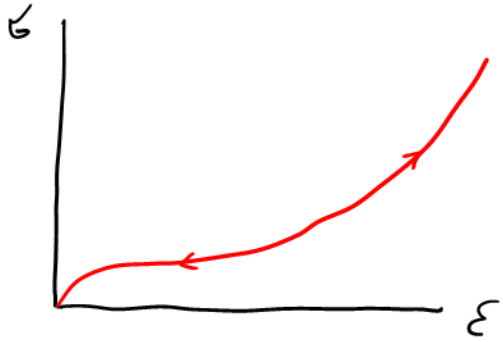
Classes of material response

Linear elasticity



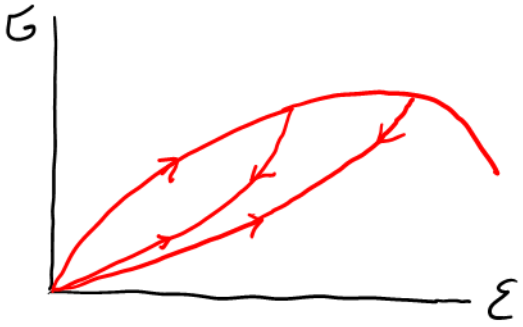
e.g. concrete/steel at small strains...

Nonlinear elasticity



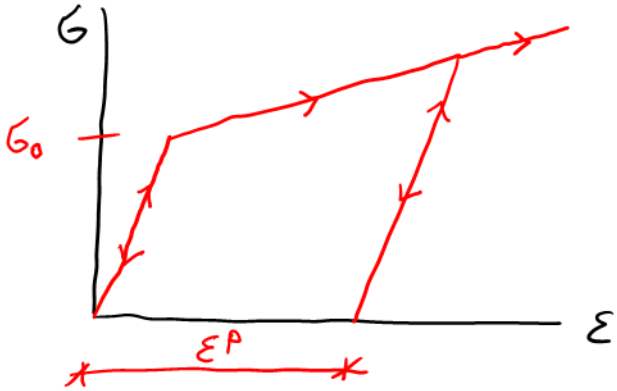
e.g. rubber...

Damage



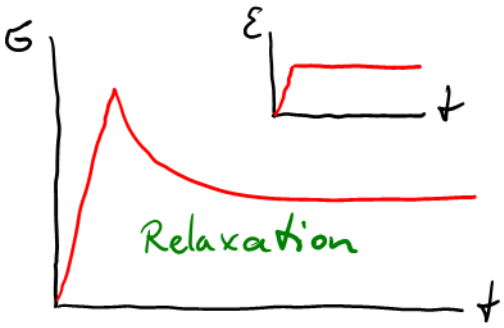
e.g. polymers...

Elasto-Plasticity



e.g. metals, soils

Visco-Elasticity



e.g. polymers...

